

Uni-modal 3D Encoders (frozen)

**3D Object
(Objaverse)**

PointNet

1024-dim

SparseConv

512-dim[†]

3D feature matrix

$$\mathbf{X} \in R^{n \times p}$$

Text Encoders (frozen)

**Text Caption
(Cap3D)**

OpenCLIP ViT-bigG-14

1280-dim

RoBERTa

768-dim

BERT

768-dim

Text feature matrix

$$\mathbf{Y} \in R^{n \times q}$$

Canonical Correlation Analysis

$n_A = 30,000$ anchor pairs
 $k = 50$ subspace dimensions

Shared Subspace $\mathbf{X}^r, \mathbf{Y}^r \in R^{n \times 50}$

Alignment Method

*CCA+Affine / KCCA
/ RML / GW variants*

Retrieval Evaluation

*Match / Top-1 / Top-5 /
Top-10 (n_q = 500 queries)*

[†]SparseConv uses random weight initialisation (pre-trained weights requested from authors, not yet available).